



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2023.– 2024 (Odd Semester)

Name of the Faculty	:	Dr. J.Jerold John Britto AP(Sr.Gr.)/Mech
Course Code and Title	:	CME340 CAD/CAM (Honors Course)
Year and Branch	:	III -Mechanical
Unit/ Topic	:	Unit-V CAM-CNC Part Programming
Course Outcome	:	Develop the computer aided machining and writing part programming.
Topic Learning Outcome	:	Develop and optimize CNC part programs, demonstrating a comprehensive understanding of machining principles, toolpath generation, and utilization of computer-aided design (CAD) and computer-aided manufacturing (CAM) software.
Activity Chosen	:	CNC Mastery: From Code to Cut
Time Allotted for Activity	:	50 minutes
Date	:	30.10.2023
No.of Students participated	:	10
Justification	:	The development of computer-aided machining and writing CNC part programming, along with a hands-on demonstration on CNC machines such as the MTAB Flexturn and FlexMill, serves as a crucial initiative in bridging the gap between theoretical knowledge and practical skills in the field of manufacturing. This activity provides a comprehensive understanding of the working principles of CNC milling and turning, coupled with in-depth insight into G-code and M-code programming. Through practical demonstrations, participants can grasp the intricacies of input and

		output devices, gaining essential experience in basic CNC machine operations. This initiative is justified as it empowers individuals with practical skills, ensuring they are well-equipped to operate and program CNC machines, ultimately enhancing their capabilities and employability in the competitive manufacturing industry.
Details of Implementation	:	Session Overview (5 minutes): • Introduction to CNC (Computer Numerical Control) machining and its significance in modern manufacturing. • Brief overview of MTAB Flexturn (CNC Turning) and FlexMill (CNC Milling) machines. • Importance of G code and M code in CNC programming. Introduction to CNC Part Programming (10 minutes): • Definition and role of CNC part programming. • Explanation of G code and M code: G code for motion commands and M code for miscellaneous functions. • Overview of the CNC programming process. Understanding MTAB Flexturn & FlexMill (10 minutes): • Detailed explanation of MTAB Flexturn and FlexMill machines. • Key features, specifications, and capabilities of each machine. • Demonstration of the working principle for CNC milling and turning operations. G Code and M Code Understanding (10 minutes): • In-depth discussion on G code and its applications in CNC machining.

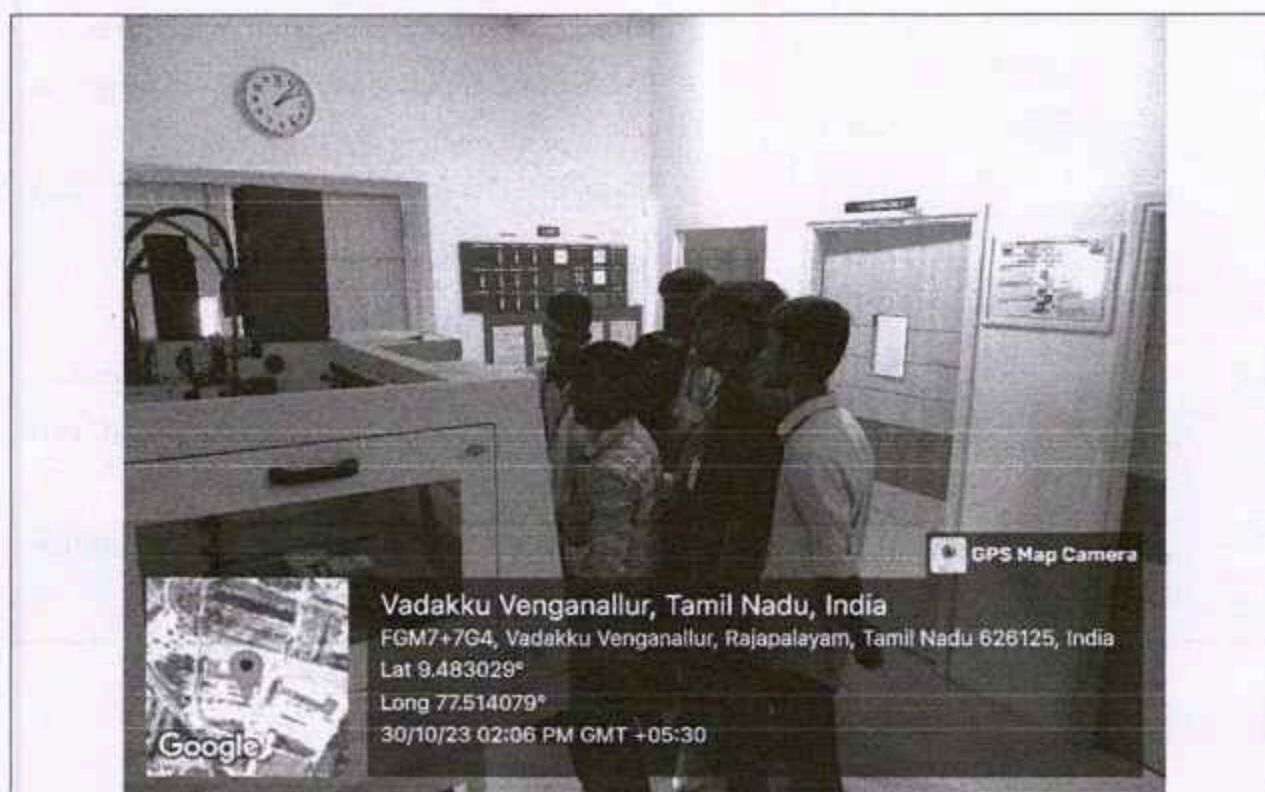
		• Explanation of common G codes used in CNC programming. • Introduction to M code and its significance. Input and Output Devices (5 minutes): • Overview of input devices used in CNC machines. • Explanation of output devices and their role in CNC machining. • Importance of accurate data input for successful CNC operations. Basic Operation Demo (5 minutes): • Hands-on demonstration of basic CNC operations on MTAB Flexturn and FlexMill machines. • Executing a simple CNC program to showcase the practical application of G code and M code. • Discussion on the importance of precision and accuracy in CNC machining. Q&A and Troubleshooting (5 minutes): • Open floor for questions from participants. • Addressing common issues and challenges in CNC programming and operation. • Providing tips for troubleshooting and optimizing CNC programs. Conclusion and Recap (5 minutes): • Summary of key points covered in the session. • Emphasis on the practical aspects of CNC part programming and machine operation. • Encouraging participants to explore further resources for continuous learning.
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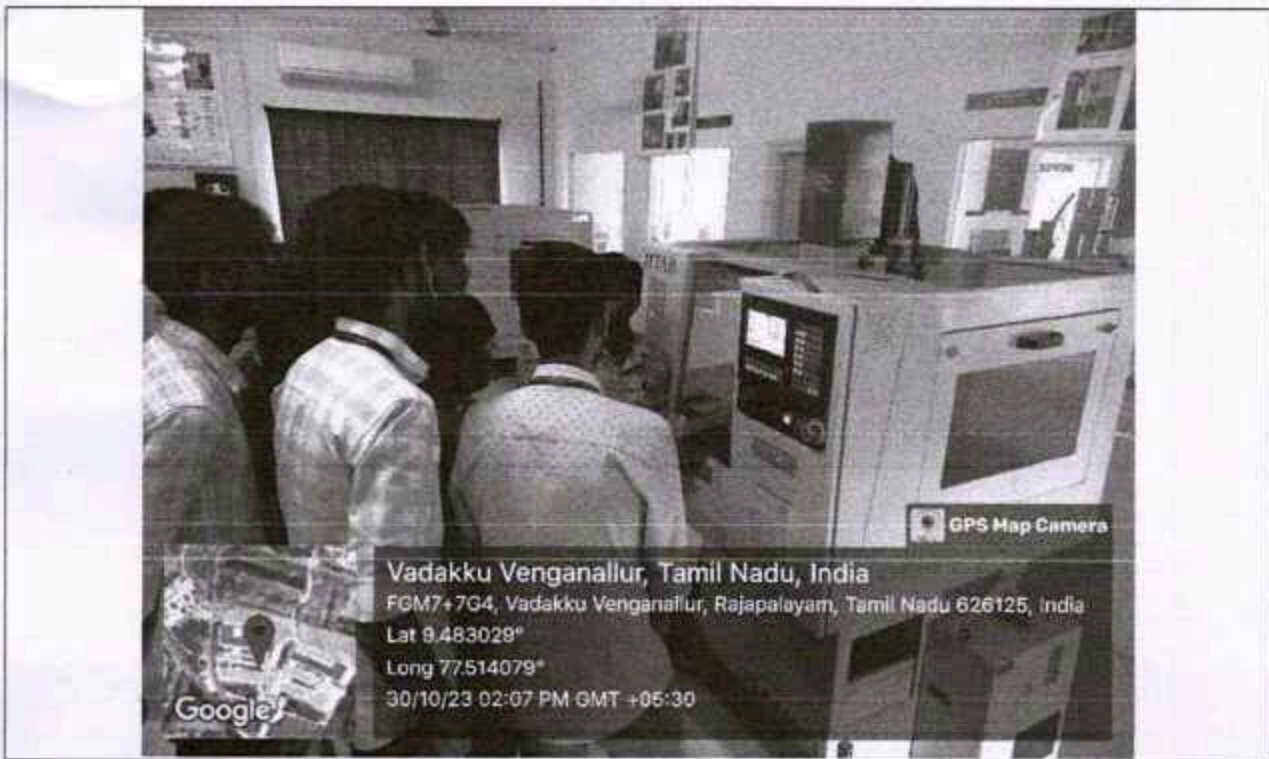
PO/PSO mapping for the activity

Course Outcome /Programme Outcome	PO1	PO3	PO5	P12	PSO1
Level of Mapping	2	2	3	2	2
Justification for correlation	Understand the principles of CNC machining and part programming for MTAB Flexturn & FlexMill.	Develop solutions for CNC machining tasks using MTAB Flexturn & FlexMill.	Utilize computer-aided machining tools for writing and optimizing CNC part programs.	Recognize the need for continuous learning in the rapidly evolving field of CNC machining.	To design component using CAD package

(1 – Low, 2- Moderate, 3-High)

Activity Glimpses





Reflective Critique:

i. *Feedback of practice from students and other stakeholders:*

Students gave positive oral feedback

ii. *Benefit of the practice:*

Students learnt to use CAD software package and file format topics.

iii. *Whether the practice is adopted in any of the courses early:*

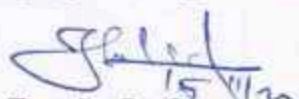
Yes, in the same course

Challenges faced in implementation:

Implementing computer-aided machining and writing part programming, particularly with CNC machines like MTAB Flexturn and FlexMill, presents challenges in seamlessly integrating theoretical knowledge with practical application. Ensuring a comprehensive understanding of G code and M code, along with proficiency in input and output devices, demands meticulous training. The hands-on demo poses logistical challenges, requiring precise coordination between theoretical concepts, machinery setup, and real-time execution to achieve effective learning outcomes. Balancing these elements within the constraints of a single session necessitates strategic planning and instructional clarity.

References: Nil

CO2: Students will be able to Develop the computer aided machining and writing part programming.


Faculty In Charge


HoD/Mech